SECTIONS OF A SHEAF ON A SPECTRAL SPACE AGREE WITH SECTIONS ON A CLOSED SUBSET MEETING EVERY CLOSURE CONNECTEDLY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the injectivity argument by specialisation, the two reductions (filtered colimits and the embedding), the inheritance of the hypotheses by the closed pieces, and the final constant-sheaf case.

Editorial note. The label lemma-h0-topological names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the nine cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Spectral spaces and quasi-compact opens. Topology, the definition of a spectral space and the lemma that a closed subset with the induced properties is again spectral. Supplies the setting and the inheritance of the hypotheses by the closed subsets Z_i.
- Description of a sheaf as a filtered colimit of constructible-like sheaves. Modules, the equation describing the auxiliary sheaves and the lemma that every sheaf is a filtered colimit of them. Supplies the reduction to a sheaf of the auxiliary form via commutation with filtered colimits.
- Embedding into a finite product of pushforwards of constant sheaves. Modules, the lemma embedding a sheaf of the auxiliary form into a finite product of (Z_i -> X)_* of constant sheaves. Supplies the reduction to the case of a finite constant sheaf.
- The connectedness decomposition for spectral spaces. Same chapter, the lemma turning a decomposition of a closed subset with connected closures-intersections into a decomposition of the whole space. Supplies the final case of a finite constant sheaf.

Source reference. Original lemma 0038: sources/stacks-chow/tags.txt, source lines 125–125. Prerequisite pointer invoked at proof line(s) 15127; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CAM: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 15072–15072. Prerequisite pointer invoked at proof

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line(s) 15162; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0CAJ: sources/stacks-chow/tags.txt, source lines 15439–15439. Prerequisite pointer invoked at proof line(s) 15124; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CAL: sources/stacks-chow/tags.txt, source lines 15441–15441. Prerequisite pointer invoked at proof line(s) 15147; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CAI: sources/stacks-chow/tags.txt, source lines 15438–15438. Prerequisite pointer invoked at proof line(s) 15125; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 009F: sources/stacks-chow/tags.txt, source lines 351–351. Prerequisite pointer invoked at proof line(s) 15129; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 008K: sources/stacks-chow/tags.txt, source lines 320–320. Prerequisite pointer invoked at proof line(s) 15128; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 08YG: sources/stacks-chow/tags.txt, source lines 11339–11339. Prerequisite pointer invoked at proof line(s) 15104; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0902: sources/stacks-chow/tags.txt, source lines 11395–11395. Prerequisite pointer invoked at proof line(s) 15160; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $Z \subset X$ be a closed subset of a topological space X. Assume*

(1) *X is a spectral space (Topology, Definition 08YG), and*
(2) *for $x \in X$ the intersection $Z \cap \overline{\{x\}}$ is connected (in particular nonempty).*

Then for any sheaf $\mathcal{F}$ on X we have $\Gamma(X, \mathcal{F}) = \Gamma(Z, \mathcal{F}|_Z)$.

Proof. If $x \rightsquigarrow x'$ is a specialization of points, then there is a canonical map $\mathcal{F}_{x'} \to \mathcal{F}_x$ compatible with sections over opens and functorial in $\mathcal{F}$. Since every point of X specializes to a point of Z it follows that $\Gamma(X, \mathcal{F}) \to \Gamma(Z, \mathcal{F}|_Z)$ is injective. The difficult part is to show that it is surjective.

Denote $\mathcal{B}$ be the set of all quasi-compact opens of X. Write $\mathcal{F}$ as a filtered colimit $\mathcal{F} = \operatorname{colim} \mathcal{F}_i$ where each $\mathcal{F}_i$ is as in Modules, Equation (0CAJ). See Modules, Lemma 0CAI. Then $\mathcal{F}|_Z = \operatorname{colim} \mathcal{F}_i|_Z$ as restriction to Z is a left adjoint (Categories, Lemma 0038 and Sheaves, Lemma 008K). By Sheaves, Lemma 009F the functors $\Gamma(X, -)$ and $\Gamma(Z, -)$ commute with filtered colimits. Hence we may assume our sheaf $\mathcal{F}$ is as in Modules, Equation (0CAJ).

Suppose that we have an embedding $\mathcal{F} \subset \mathcal{G}$. Then we have

$$\Gamma(X, \mathcal{F}) = \Gamma(Z, \mathcal{F}|_Z) \cap \Gamma(X, \mathcal{G})$$

where the intersection takes place in $\Gamma(Z, \mathcal{G}|_Z)$. This follows from the first remark of the proof because we can check whether a global section of $\mathcal{G}$ is in $\mathcal{F}$ by looking at the stalks and because every point of X specializes to a point of Z.

By Modules, Lemma 0CAL there is an injection $\mathcal{F} \to \prod (Z_i \to X)_* \underline{S_i}$ where the product is finite, $Z_i \subset X$ is closed, and S_i is finite. Thus it suffices to prove surjectivity for the sheaves $(Z_i \to X)_* \underline{S_i}$. Observe that

$$\Gamma(X, (Z_i \to X)_* \underline{S_i}) = \Gamma(Z_i, \underline{S_i}) \quad \text{and} \quad \Gamma(X, (Z_i \to X)_* \underline{S_i}|_Z) = \Gamma(Z \cap Z_i, \underline{S_i})$$

Moreover, conditions (1) and (2) are inherited by Z_i; this is clear for (2) and follows from Topology, Lemma 0902 for (1). Thus it suffices to prove the lemma in the case of a (finite) constant sheaf. This case is a restatement of Lemma 0CAM which finishes the proof. $\square$